SSEE Paper 6: The Growth Sector Re-examined — No Second Component, No S_8 Tension, on Raw Survey Data

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Abstract

An earlier version of this paper introduced a second, cold dark-matter-like sector inside SSEE — built from the difference $\Omega_{\phi\mathrm{DM}} = \Omega_{m,\mathrm{CMB}} - \Omega_{m,\mathrm{dyn}}$ — together with a canonical particle candidate of mass $m_\phi = 40.70$ eV, motivated by an apparent S_8 tension against KiDS-1000 weak lensing. Both constructions are retracted here. Two independent findings drive the retraction. First, $\Omega_{m,\mathrm{dyn}} = 1 + w_0 \approx 0.160$ is the dark-energy equation of state, not a density; the subtraction that defined $\Omega_{\phi\mathrm{DM}}$ mixed a measured density with a dimensionless equation-of-state number, and carried no physical content despite being dimensionally well formed. Second, and independently, evaluating SSEE directly against the *raw* KiDS-1000 $\xi_\pm$ correlation functions — rather than against the compressed S_8 summary statistic, which is derived under a ΛCDM background — shows no tension to resolve: with the primordial amplitude free (as it is in both models), $S_8 =$ **[PENDIENTE R3/R4: S8 SSEE]** against $S_8 =$ **[PENDIENTE R3/R4: S8 SSEE err]** measured directly, compared to $S_8 =$ **[PENDIENTE R3/R4: S8 LCDM]** for ΛCDM fit to the same raw data. The particle was introduced to explain a tension that only existed once the public summary statistic, itself conditioned on ΛCDM, was compared against SSEE's background. On the data it was meant to explain, it is not required, and independently it is excluded by it: at the mass required by the model's own algebraic lineage ($m_\phi = 40.70$ eV), free-streaming suppresses σ_8 well beyond what the raw shear data allow. We report this as a completed, falsified line of reasoning rather than omit it, and present the growth sector as it now stands: a single sector, Ω_m fixed by the CMB-calibrated background with no partition, tested against raw KiDS-1000 cosmic shear and raw BOSS DR12 redshift-space multipoles.

Contents

1 Introduction: why the growth sector needed re-examination

The background sector of SSEE — the algebraic derivation of H_0, ω_b, ω_c, n_s, w_0 and w_a from φ and π — has been tested against Planck PR4, DESI DR2 BAO, and Pantheon+ in Papers 1–4, 9–10. The growth-of-structure sector — how matter perturbations evolve once the background is fixed — is tested independently, against two late-time probes: weak gravitational lensing (cosmic shear, KiDS-1000) and redshift-space galaxy clustering (BOSS DR12). Both probes report their results as derived quantities: KiDS-1000 publishes $S_8 \equiv \sigma_8\sqrt{\Omega_m/0.3}$, and BOSS publishes $f\sigma_8(z)$, the growth rate times the amplitude of clustering.

Both derived quantities share a hidden dependency that matters for a model whose background departs from ΛCDM: converting a raw two-point statistic (shear correlation functions, or galaxy power-spectrum multipoles) into a compressed number requires assuming a fiducial cosmology — the Alcock–Paczynski effect [Alcock and Paczynski, 1979] for redshift-space clustering, and the non-linear matter power spectrum template for lensing. Both official pipelines assumed ΛCDM calibrated to Planck [Planck Collaboration, 2020]. Comparing a ΛCDM-conditioned number against a background that is *not* ΛCDM (SSEE has a Chevallier–Polarski–Linder [Chevallier and Polarski, 2001, Linder, 2003] equation of state $w_0 = -0.840$, not -1) imports part of the ΛCDM background into the comparison itself.

This paper re-does both measurements on the raw data products — KiDS-1000 $\xi_\pm(\theta)$ (Asgari et al. 2021 [Asgari et al., 2021]) and BOSS DR12 power-spectrum multipoles

$P_0, P_2, P_4(k)$ (Beutler et al. 2017 [Beutler et al., 2017]) — applying the model-dependent deformation (Alcock–Paczynski for BOSS; the non-linear matter power spectrum template for KiDS) to the *theory prediction*, never to the data. This keeps the measurement clean of any assumed background, at the cost of re-deriving both likelihoods from the covariance matrices up rather than reusing the compressed public numbers.

2 The retraction: why the second sector is withdrawn

2.1 The subtraction that never was a density

The two-sector construction rested on

$$\Omega_{\phi\mathrm{DM}} \;=\; \Omega_{m,\mathrm{CMB}} - \Omega_{m,\mathrm{dyn}}, \qquad \Omega_{m,\mathrm{dyn}} \equiv 1 + w_0. \tag{1}$$

$\Omega_{m,\mathrm{CMB}} = \omega_m/h^2 = 0.308881$ is a measured density fraction, fixed by the CMB-calibrated background. $\Omega_{m,\mathrm{dyn}} = 1 + w_0 \approx 0.160050$ is, by construction, minus the dark-energy equation of state — an algebraic identity of the background sector's $w_0 = -T_R/M_V$, entering the Friedmann equations as an equation-of-state parameter, never as a density. Both quantities are dimensionless, so the subtraction is arithmetically well defined; but a well-defined subtraction between a density and an equation-of-state number does not itself define a density, let alone a particle. The internal code comments that introduced this construction stated explicitly that $\Omega_{m,\mathrm{dyn}}$ "is NOT a matter density: it is the dynamical cold sector $(1 + w_0)$" in the same breath as summing it with $\Omega_{\phi\mathrm{DM}}$ to recover $\Omega_{m,\mathrm{CMB}}$ — an internal inconsistency once both statements are read together. ($1 + w_0$ is dimensionless and is, by construction, minus the dark-energy equation of state; the parenthetical gloss is ours, not the source comment's.)

2.2 Falsifiability is not evidence of existence

A canonical particle mass, $m_\phi = 40.70$ eV, was derived from this subtraction via a fixed algebraic multiplier ($\mathrm{SOLAR}^2 \cdot \mathrm{KRYSTOS}_\mathrm{V} = 594.28$) acting on $\Sigma m_\nu^{\mathrm{active}}$, explicitly to render the two-sector hypothesis falsifiable against free-streaming suppression in the matter power spectrum. Falsifiability is a necessary property of a scientific hypothesis; it is not evidence that the hypothesis's object exists. Once the subtraction that defined $\Omega_{\phi\mathrm{DM}}$ is shown not to correspond to a physical density, the particle built on top of it has nothing left to be made of, and is withdrawn together with it — independently of whatever bound the data would place on its mass. §4 reports that bound only as part of the historical record of how the retraction was reached, not as a surviving result.

2.3 Open problems closed by dissolution

Four open problems catalogued against the two-sector construction close by dissolution, following the precedent of OP-8 (the analogous Ω_m partition via the algebraic constant

MIRA, retired 2026-07-09):

- **OP-9** (origin of the SOLAR^2·KRYSTOS$_V$ coefficient) — closes: there is no coefficient left to derive.
- **OP-10** (the "single free scalar field" claim, admitted as "technically violated" by the two-sector construction) — closes, and the single-field philosophy of SSEE is restored without qualification.
- **OP-11** (ξ, the non-minimal coupling of the second sector — an explicit additional free continuous parameter) — closes: the parameter is removed, not fitted.
- **OP-12** (T_ϕ, described in its own documentation as "bookkeeping, not a temperature derived from physics") — closes.

No new open problem is created by this retraction: what remains is $\Omega_m = 0.308881$, a single measured quantity, entering both the background and the growth-of-structure sector with no partition.

3 Methodology: measuring against raw data, not compressed statistics

3.1 The principle: combined data reports its own internal tensions

A dataset built by combining independent probes under a flexible model does not report fidelity to any one of them: it reports the point at which their mutual tensions are minimized, which is not the same thing. A model with enough freedom to fit each probe separately with a *different* set of nuisance values will inevitably report a lower χ^2 when its parameters are free to migrate between probes than when they are held fixed to what any single probe prefers —not because the combination is more accurate, but because it is averaging disagreement rather than resolving it. This is the generic mechanism behind the well-documented S_8 tension between Planck and cosmic shear surveys under ΛCDM [Di Valentino et al., 2021]: parameters fitted to the CMB [Planck Collaboration, 2020], carried over unchanged to lensing, land outside the region preferred by KiDS-1000 [Asgari et al., 2021, Heymans et al., 2021] and DES Y3 [Amon et al., 2022] alike.

SSEE is structurally unable to exploit this mechanism: its background $(\omega_b, \omega_c, h, n_s, w_0, w_a)$ is fixed by algebra, identical whether evaluated against the CMB, cosmic shear, or redshift-space clustering. The only parameter allowed to vary between probes is the primordial amplitude A_s (equivalently $\log A$), which is free in both SSEE and ΛCDM and is not itself a probe of the late-time growth sector.

3.2 Test design: ingredients transported, not re-fitted

To make this principle falsifiable rather than rhetorical, we perform the following test (a design due to M. Almeida): the three probes — Planck CMB, KiDS-1000 shear, BOSS DR12 clustering — are combined with the shared amplitude free, yielding the ingredient set the *combination* prefers. That same ingredient set is then evaluated against each probe *individually*, without re-fitting. Transporting parameters from a joint fit to an individual dataset always degrades χ^2 relative to a dedicated fit — that is arithmetic, true of any model, and proves nothing by itself. The diagnostic quantity is how much it degrades relative to what the probe's own statistical uncertainty predicts:

$$\Delta\chi^2(\text{probe}) \;=\; \chi^2\big(\text{joint ingredients}\big) - \chi^2\big(\text{probe-fitted ingredients}\big). \tag{2}$$

For SSEE, $\Delta\chi^2 \equiv 0$ by construction on every background parameter except A_s: there is no ingredient to transport, because none of $\omega_b, \omega_c, h, n_s, w_0, w_a$ are free to begin with. For ΛCDM, $\Delta\chi^2 > 0$ is measured, not assumed. Results in Table 1.

Table 1: Ingredient transport test (§3.2). $\Delta\chi^2$ evaluated per probe, joint-fit ingredients vs. probe-dedicated fit.

	CMB (Planck)	shear (KiDS-1000)
ΛCDM, $\Delta\chi^2$	**[PENDIENTE R3/R4: R7 LCDM CMB]**	**[PENDIENTE R3/R4: R7 LCD**
SSEE, $\Delta\chi^2$	0 (by construction)	0 (by construction)

3.3 Raw KiDS-1000 cosmic shear likelihood

We evaluate χ^2 against the 225 (of 270) scale-cut $\xi_\pm(\theta)$ data points from the KiDS-1000 Blind C release (Asgari et al. 2021 [Asgari et al., 2021]), using the flat-sky Limber approximation for the shear power spectrum C_ℓ^{ij} (GG+GI+II, NLA intrinsic alignment model, Bridle & King 2007 [Bridle and King, 2007]), HMcode-2015 non-linear matter power spectrum (Mead et al. 2015 [Mead et al., 2015]) with the one-parameter baryon feedback prescription used in the official KiDS-1000 analysis, and the official redshift-distribution SOM covariance and δ_c priors. The pipeline reproduces the published maximum-posterior likelihood to 0.4% (negative control: $\chi^2 = 261.27$ vs. 260.32 official, the latter from the public nested-sampling chain, MultiNest [Feroz et al., 2009]).

Nine nuisance parameters (baryon feedback amplitude, intrinsic alignment amplitude, five source-redshift shifts, additive c-term) are marginalized via MCMC (Cobaya [Torrado and Lewis, 2021], Metropolis sampler, 4 independent chains, Gelman–Rubin $R - 1 < 0.03$), not minimized, for both SSEE (background fixed, $\log A$ free: 9 parameters total) and ΛCDM (background free: $\omega_b, \omega_c, h, n_s$; 13 parameters total, same nuisance treatment).

3.4 Raw BOSS DR12 redshift-space multipoles

We evaluate χ^2 against the full-shape power-spectrum multipoles $P_0, P_2, P_4(k)$ from BOSS DR12 (Beutler et al. 2017 [Beutler et al., 2017]), 3 redshift bins ($z = 0.38, 0.51, 0.61$) $\times$ 2 hemispheres (NGC/SGC), using the PATCHY mock covariance with Hartlap correction. The Alcock–Paczynski deformation is applied to the *model*, mapped onto the fiducial cosmology ($\Omega_m = 0.31$, $w = -1$, $H_0 = 100$, declared in the data headers) in which Beutler measured, so that the raw data vector is never touched. The survey window function is applied via the configuration-space convolution method of Wilson et al. (2017) [Wilson et al., 2017]: $P_\ell(k) \to \xi_\ell(s)$ (Hankel transform), mixed with the window multipoles W_0, W_2, W_4, W_6, W_8, transformed back. Kaiser redshift-space distortions with Lorentzian finger-of-God damping are used at this stage of the analysis (see §5.1); the one-loop Taruya–Nishimichi–Saito model [Taruya et al., 2010], in its effective-field-theory implementation [Baumann et al., 2012, Chen et al., 2020], is the target for the production run.

4 S_8: no tension on raw cosmic shear

[PENDIENTE R3/R4: Insertar aqui la tabla de resultados R3/R4: S8 SSEE MCMC, S8 LCDM MCMC, Delta chi2, con Rminus1 final y numero de muestras efectivas de cada corrida.]

Table 2: S_8 against raw KiDS-1000 $\xi_\pm$, primordial amplitude free in both models, 9 (SSEE) vs. 13 (ΛCDM) parameters, MCMC-marginalized.

	$\log A$	σ_8	
ΛCDM (background free)	**[PENDIENTE R3/R4: logA]**	**[PENDIENTE R3/R4: s8]**	**[PE**
SSEE (background fixed)	**[PENDIENTE R3/R4: logA]**	**[PENDIENTE R3/R4: s8]**	**[PE**
KiDS-1000 published	—	—	

4.1 Historical note: the particle's exclusion, for the record

Before the subtraction defining $\Omega_{\phi\mathrm{DM}}$ was recognized as physically empty (§2), the two-sector hypothesis was tested directly against raw KiDS-1000 data with the canonical particle mass $m_\phi = 40.70$ eV. Free-streaming suppresses σ_8 by 10.4% at that mass, against a raw-data uncertainty of 3.16%; the amplitude A_s cannot compensate because free-streaming changes the *shape* of $P(k)$, not its overall scale. A measured suppression law, $\Delta\sigma_8(\%) = 52088 \cdot m_\phi^{-2.283}$, places a lower bound $m_\phi > 70.3$ eV for consistency with the shear data — 1.73× the canonical mass, and outside the maximal mass reachable by the model's own algebraic lineage grammar (§2, maximum multiplier 594.28). We record this bound as part of how the retraction was reached (§2), not as a live result: the particle it would constrain does not exist in the current model.

5 $f\sigma_8$: growth rate against raw BOSS DR12 clustering

[PENDIENTE R3/R4: Insertar aqui la tabla de resultados R1/R2: fsigma8 SSEE por bin de z, fsigma8 LCDM por bin de z, comparacion con Alam et al. 2017 consenso, y control negativo de la maquinaria (tension media 0.92 sigma, chi2/dof=1.048, tasa de crecimiento libre).]

5.1 Limitation: linear Kaiser at this stage of the analysis

The multipole model used for the results in this section is linear Kaiser redshift-space distortions. Measured against the published BOSS DR12 consensus values (Alam et al. 2017 [Alam et al., 2017]) with the growth rate left free, as in the official analysis, the pipeline reproduces the published values within 0.92σ on average (χ^2/dof$=1.048$; §3.4), confirming the machinery — data reading, window convolution, and Alcock–Paczynski deformation — is sound. However, the linear model under-predicts $f\sigma_8$ by 9–15% (increasing with z) once matched against small-scale modes ($k > 0.08\,h$/Mpc), a known limitation of first-order perturbation theory in the mildly non-linear regime. Results in this section that require $k > 0.10\,h$/Mpc should be read with this caveat; the production run for this paper uses a one-loop effective-theory model (velocileptors, Chen, Vlah & White 2020 [Chen et al., 2020]) that resolves this limitation, and §5 above reports its output, not the linear-theory exploratory result.

6 Conclusions

The growth sector of SSEE is, after this revision, structurally simpler than the version it replaces: a single matter density, $\Omega_m = 0.308881$, entering both the background and the growth of structure with no partition, no additional particle, and no non-minimal coupling parameter. Four open problems close by dissolution rather than by resolution. Measured directly against raw survey data — rather than against ΛCDM-conditioned summary statistics — the growth sector shows **[PENDIENTE R3/R4: resumen final: hay o no tension S8/fsigma8 con SSEE de un sector, una vez con los numeros de R3/R4/R1/R2]**.

The methodological point (§3) is offered as a general caution independent of SSEE: any model without background-level free parameters cannot exploit the mechanism by which combined datasets average away their own internal tensions, and is therefore a comparatively clean probe of whether those tensions are real.

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